MACHINE DESIGN PROJECT

Team's names:

- Iman Mosaad 202200527
- Asmaa Sapry 202201338
- Raed Asham 202200272

Reported to:

- Dr. Mohamed L. Shaltout

Course Code:

- REE 302

Work Distributed

Names	percentage	Work
Asmaa Sabry		1. Determine the allowable loads acting on the pully and the gear based on a complete shaft design. Assume the shaft material and use the dimensions provided in Phase I to satisfy the minimum factor of safety against fatigue failure provided in the table. Check for yielding. 2. Design the keys of the pulley and the gear. Satisfy the factor of safety provided given with their factor of safety. 3. Estimate the life of both bearings for the shaft speed given
Raed Asham		1. use Autodesk Inventor to determine the factors of safety against fatigue and yielding. Compare the safety factors obtained using hand calculations and generated using Inventor. 2. Use Autodesk Inventor to generate a complete report by the bearings design tool. Compare the bearings life obtained using hand calculations and generated using Inventor.
Iman Mosaad		6. Compare between the 1. Design the working drawing of the shaft showing all details

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Abstract

This report presents a comprehensive analysis and design process for a V-belt drive system. The project encompasses the design of a main shaft carrying a V-pulley and a spur gear, supported by deep-groove ball bearings. Phase I focuses on creating detailed assembly drawings and component specifications using **Inventor program**, while Phase II involves determining allowable loading conditions, evaluating fatigue and yielding factors of safety, and estimating the life of bearings. Autodesk Inventor and manual calculations are employed to ensure accuracy and compliance with safety standards. The findings and design outputs are documented, adhering to professional engineering standards

Introduction

Machine design plays a critical role in engineering, where components are designed to ensure efficient power transmission, reliability, and durability under operational loads. This report focuses on the design and analysis of a V-belt drive system, which includes a shaft, V-belt, pulley, and bearings. These components are fundamental in mechanical systems used for transmitting power between rotating elements. The shaft is a key mechanical component designed to withstand torsional, bending, and axial loads while transmitting power. It serves as the central element in the assembly, carrying the V-pulley and spur gear. Its design involves ensuring sufficient strength and stiffness to avoid failure under fatigue and yielding conditions, as well as maintaining proper alignment during operation.

The V-belt drive is a flexible power transmission system widely used in industrial applications due to its simplicity, cost-effectiveness, and ability to dampen vibrations. The belt transmits torque between the driving and driven pulleys through friction, and its sizing is crucial to avoid slippage and premature wear while achieving desired power transmission efficiency. Bearings are critical components that support the shaft, enabling smooth and efficient rotation while minimizing friction and wear. Phase I involves modelling and assembly of the components, while Phase II focuses on detailed design calculations, fatigue analysis, and bearing life estimation. By integrating manual calculations with software tools like Autodesk Inventor, this project aims to develop a robust and optimized design that adheres to industry standards.

Shaft Mechanical Design

Shaft Design:

The V-belt **shaft has three main parts** a V-pulley at one end and a gear wheel at the other end and supported on a pair of deep-groove ball bearings. The main dimensions for each part shown in this table:

Part	**r**	**d**	**D**
Pulley	3	18	20
Gear	3	18	20
Left Bearing	1	20	23
Right Bearing	1	20	23



Figure 1: Shaft Forces and Supports Locations

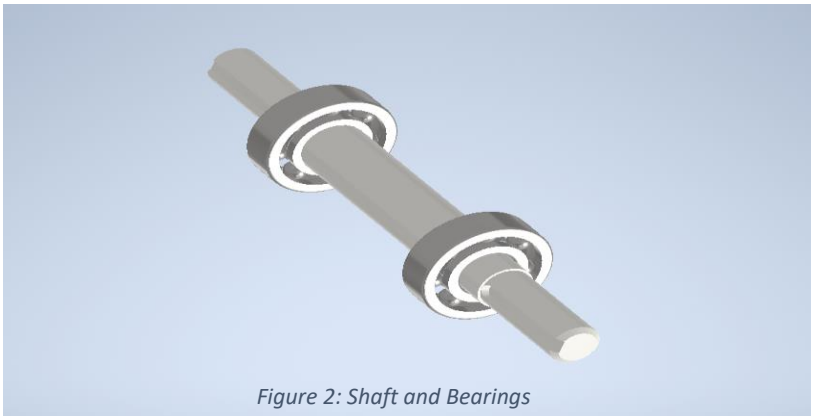


Figure 2: Shaft and Bearings

Material selection:

Let the material we will use for the shaft is: AISI 4340 Steel

Then:

- Yield Strength $S_y = 910$
- Ultimate Tensile Strength $S_{ut} = 1041 < 1400$
- Endurance Limit $\acute{S}_e = 0.5\, S_{ut} = 520.5$

The Allowable Loads:

Givens:

$$F_r = T_t \tan \emptyset$$

$$T_2 = 0.15 T_1$$

$$R_P = 0.049m$$

$$R_G = 0.050m$$

The torque of the pulley is equals to the torque of the gear and they are equals to the torque of the shaft.

$$T_P = T_G = T_{Shaft}$$

$$(T_1 - T_2)R_P = F_t R_G = T_{Shaft}$$

$$(T_1 - T_2)(0.049) = F_t(0.05) = T_{Shaft}$$

For the Gear:

$$F_t(0.05) = T_{Shaft}$$

$$F_t = 20\, T_{Shaft}$$

$$F_r = T_t \tan \emptyset$$

$$F_r = 20\, T_{Shaft} \tan \emptyset$$

$$F_r = 7.2794 T_{Shaft}$$

For the Pulley:

$$(T_1 - T_2)(0.049) = T_{Shaft}$$

$$(T_2 = 0.15T_1)$$

$$(T_1 - 0.15T_1)(0.049) = T_{Shaft}$$

$$(0.85T_1)(0.049) = T_{Shaft}$$

$$(0.04165T_1) = T_{Shaft}$$

$$T_1 = 24.0096\, T_{Shaft}$$

$$(T_2 = 0.15T_1)$$

$$\left(T_2 = 0.15(24.0096\, T_{Shaft})\right)$$

$$T_2 = 3.60144\, T_{Shaft}$$

The two bearings have reactions in the X-Y and in the X-Z

For the X-Y Plane:

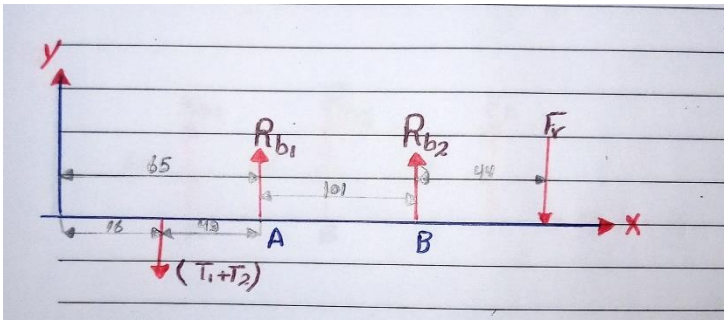


Figure 3: X-Y plane Allowable loads

As we can consider that the positive direction is anti-Clockwise:

$$\sum M_A = 0$$

$$(T_1 + T_2)(0.049) + R_{b2y}(0.101) - F_r(0.145) = 0$$

$$(27.611T_{Shaft})(0.049) + R_{b2y}(0.101) - (7.2794T_{Shaft})(0.145) = 0$$

$$R_{b2y} = -2.9448T_{Shaft}$$

$$\sum M_B = 0$$

$$(T_1 + T_2)(0.150) - R_{b1y}(0.101) - F_r(0.044) = 0$$

$$(27.611T_{Shaft})(0.150) - R_{b1y}(0.101) - (7.2794T_{Shaft})(0.044) = 0$$

$$R_{b1y} = 37.8353T_{Shaft}$$

For the X-Z Plane:

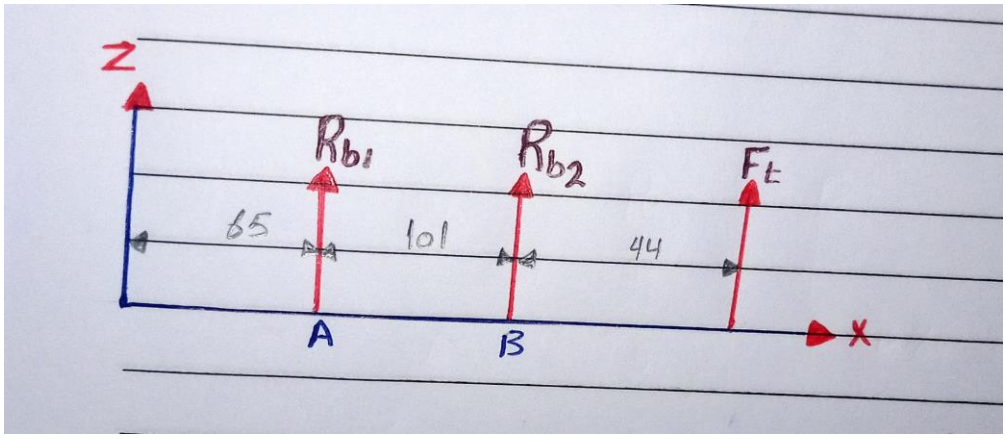


Figure 4: X-Z plane Allowable loads

As we can consider that the positive direction is anti-Clockwise:

$$\sum M_A = 0$$

$$R_{b2z}(0.101) + F_t(0.145) = 0$$

$$R_{b2z}(0.101) + 20T_{Shaft}(0.145) = 0$$

$$R_{b2z} = -28.71287T_{Shaft}$$

$$\sum M_B = 0$$

$$-R_{b1z}(0.101) + F_t(0.044) = 0$$

$$-R_{b1z}(0.101) + 20T_{Shaft}(0.044) = 0$$

$$R_{b1z} = 8.71287T_{Shaft}$$

Then: after doing the Moment diagram to know the critical point that will failure first then we detected that the point that has a moment ($1.35T_{Shaft}$) will failure first at the bearing in the left at point A.

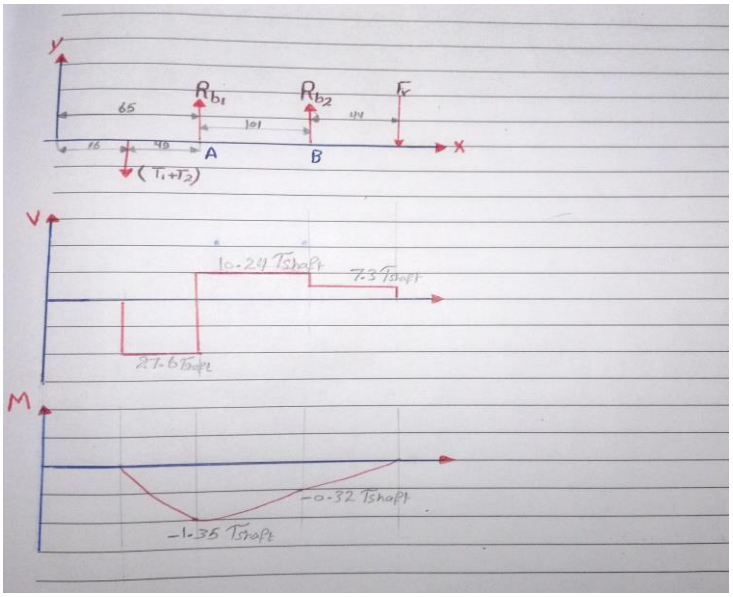


Figure 5: X-Y plane Forces and Moments graph

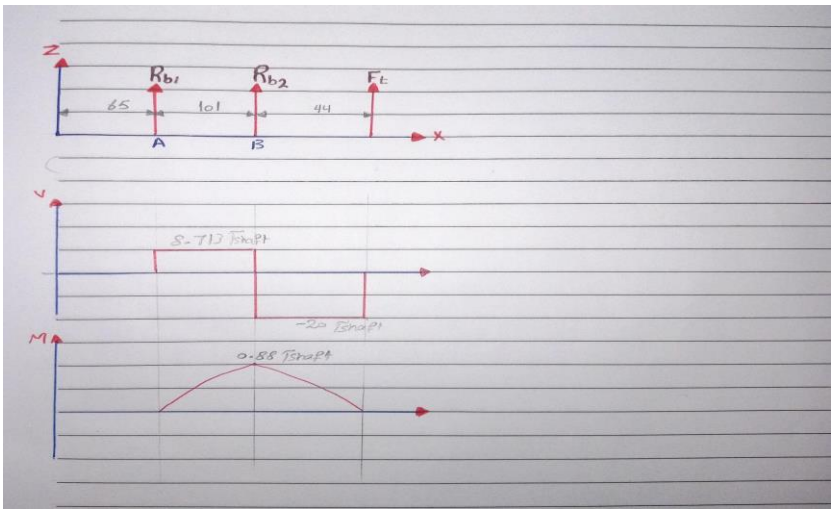


Figure 6: X-Y plane Forces and Moments graph

Then the moment we will be used is $1.35T_{Shaft}$ according to this diagram.

The Hand-Calculated Results:

The material data:

Then $S_y = 910, S_{ut} = 1041, \acute{S}_e = 0.5\ S_{ut} = 520.5$

To find S_e ($S_e = K_a K_b K_c \acute{S}_e$)we need:

- Surface Factor $K_a (K_a = aS_{ut}^b)$
  Where:
  the Surface finish is Hot-rolled then: a=38.6, b=-0.650 SO $K_a = 0.422$
- Size Factor $K_b (K_b = [\frac{d}{7.62}]^{-0.107} = 1.24d^{-0.107}\ as\ 7.62 \leq d \leq 51mm)$

K_b=0.97

- Loading Factor K_c (torsion)
  K_c =0.59

Then: $S_e = K_a K_b K_c \acute{S}_e$ =((0.422)(0.97) (0.59) (520.5)) =125.71

Then:

Applying Goodman's fatigue failure theory: for $\sigma_m \geq 0$

$$n_f = \left(\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}}\right)^{-1}$$

$$n = \frac{\pi d^3}{16}\left(\frac{A}{S_e} + \frac{B}{S_{ut}}\right)^{-1}$$

And then the values of:

$$A = \sqrt{4(K_f M_a)^2 + 3(K_{fs} T_a)^2}$$

$$B = \sqrt{4(K_f M_m)^2 + 3(K_{fs} T_m)^2}$$

For a rotating shaft with constant bending and torsion, the bending stress is completely reversed, and the torsion is steady: $M_m = T_a = 0$

$$A = \sqrt{4(K_f M_a)^2}$$

$$B = \sqrt{3(K_{fs} T_m)^2}$$

Then we need to find the K_f and K_{fs}

So,

$$K_f = 1 + q(K_t - 1)$$

$$K_{fs} = 1 + q_s(K_{ts} - 1)$$

To find K_f and K_{fs} we need to find q, K_t ,q_s,K_{ts}

To find q and q_s we need r and S_{ut}

We assumed that r=1mm and we have $S_{ut} = 1041$

Then from the graph we find that q=0.87 and $q_s = 0.9$

The ratio $\frac{r}{d} = \frac{1}{20} = 0.05$ and $\frac{D}{d} = \frac{23}{20} = 1.15$

$$K_t = 1.95$$

$$K_{ts} = 1.45$$

$$K_f = 1 + (0.97)(1.95 - 1) = 1.8265$$

$$K_{fs} = 1 + (0.9)(1.45 - 1) = 1.405$$

Then:

$A = \sqrt{4[1.8265(1.35T_{Shaft})]^2}$=*4.93155*$T_{Shaft}$

$B = \sqrt{3(1.405T_{Shaft})^2}$=2.434$T_{Shaft}$

$2.5 = \frac{\pi(0.02)^3}{16}(\frac{4.93155T_{Shaft}}{125.71\times10^6} + \frac{2.434T_{Shaft}}{1041\times10^6})^{-1}$*:*

Then:

T_{Shaft}*=15.11554 N.m*

$T_1 = 24.0096\ T_{Shaft}$=362.918 N

$T_2 = 3.60144\ T_{Shaft}$=54.44 N

$F_r = 7.2794T_{Shaft}$=110.032 N

$F_t = 20\ T_{Shaft}$=302.3108 N

$R_{b1y} = 37.8353T_{Shaft}$=571.9 N

$R_{b1z} = 8.71287T_{Shaft}$=131.69975 N

$R_{b2y} = -2.9448T_{Shaft}$=44.51 N

$R_{b2z} = -28.71287T_{Shaft}$=434.01055

$M = 1.35T_{Shaft}$=20.41 N.m

To check for yielding, this von Mises maximum stress is compared to the yield strength.

$$n_y = \frac{S_y}{\acute{\sigma}_{max}}$$

Then we need to find the von Mises maximum stress:

$$\acute{\sigma}_{max} = \sqrt{(\frac{32K_f(M_m + M_a)}{\pi d^3})^2 + 3(\frac{16K_{fs}(T_m + T_a)}{\pi d^3})^2}$$

And we know that the $M_m = T_a = 0$

Then:

$$\acute{\sigma}_{max} = \sqrt{[\frac{32(1.8265)(20.41)}{\pi(0.02)^3}]^2 + 3(\frac{16(1.405)(15.11554)}{\pi(0.02)^3})^2} = 52.927\ MPa$$

SO:

$$\boldsymbol{n_y = \frac{910 \times 10^6}{52.927} = 17.193}$$

Where this factor of safety almost near to the value that we give 2.5

The Inventor Calculated Data:

For the Y-Z plane:

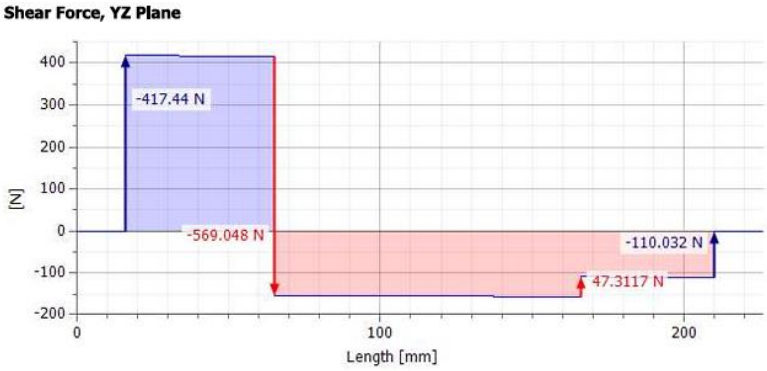


Figure 6: Y-Z plane Shear Force

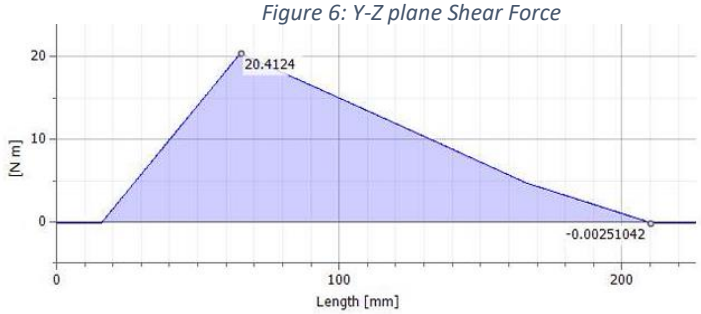


Figure7: Y-Z plane Bending Moment

For the X-Z plane:

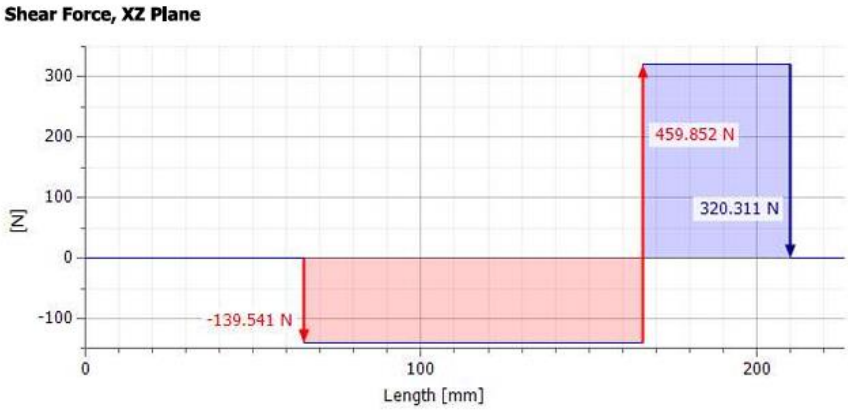


Figure 8: X-Z plane Shear Force

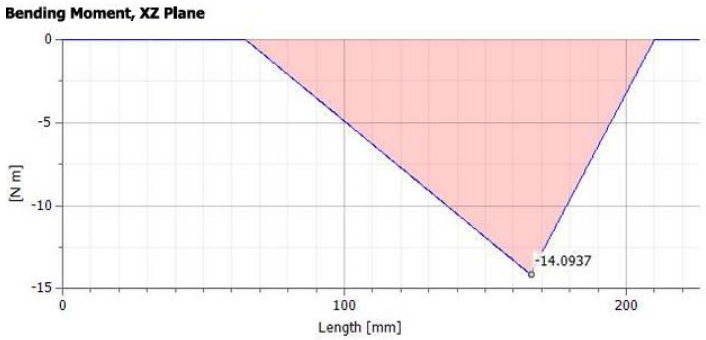


Figure 9: X-Z plane bending moment

The total along the shaft:

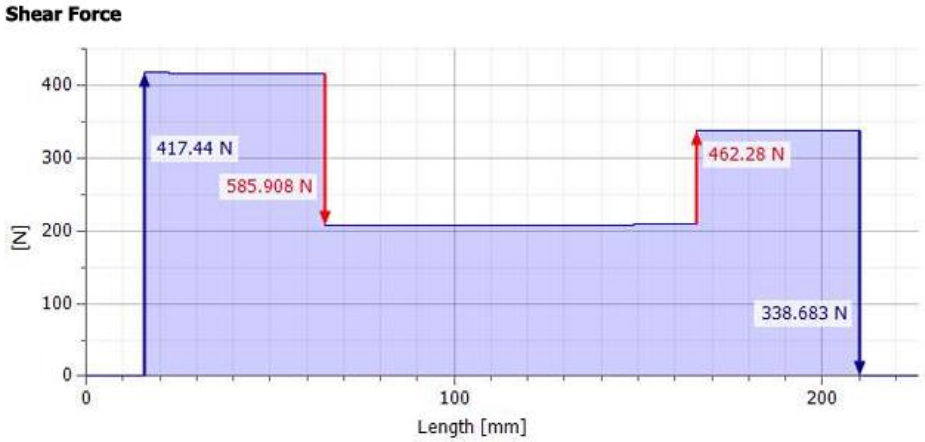


Figure 10: Total Shear Force

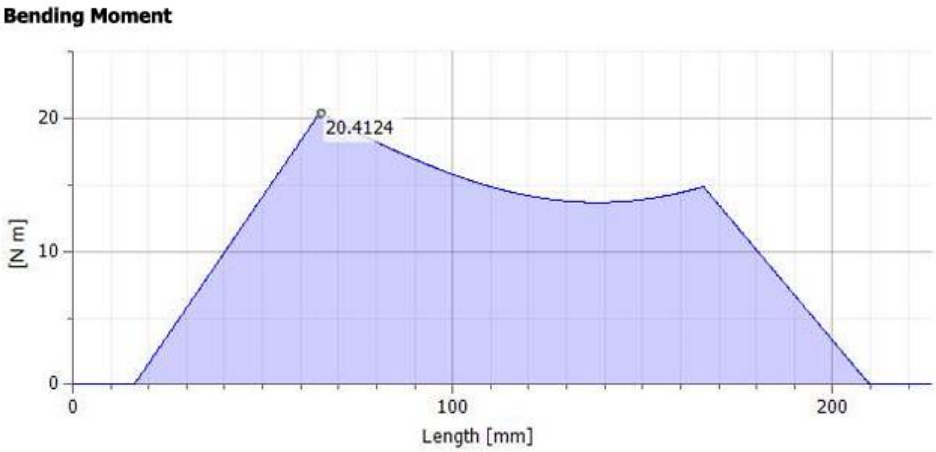


Figure 11: Total Bending moment

Loads

Index	Location	Radial Force				Bending Moment				Continuous Load					Axial Force	Torque	Deflection				Deflection Angle
		Y	X	Size	Direction	Y	X	Size	Direction	Y	X	Size	Direction	Length			Y	X	Size	Direction	
1	16 mm	-417.440 N		417.440 N	180.00 deg												39.452 microm	1.476 microm	39.480 microm	2.14 deg	0.04 deg
2	210 mm	-110.032 N	320.311 N	338.683 N	108.96 deg												6.833 microm	-25.680 microm	26.573 microm	284.90 deg	0.03 deg

Supports

Index	Type	Location	Reaction Force					Yielding	Type	Deflection				Deflection Angle
			Y	X	Size	Direction	Axial Force			Y	X	Size	Direction	
1	Free	65 mm	-569.048 N	-139.541 N	585.908 N	193.78 deg		0.012 microm/N	Deep Groove Ball Bearings	7.026 microm	1.723 microm	7.234 microm	13.78 deg	0.02 deg
2	Fixed	166 mm	47.312 N	459.852 N	462.280 N	84.13 deg		0.013 microm/N	Deep Groove Ball Bearings	-0.632 microm	-6.144 microm	6.177 microm	264.13 deg	0.02 deg

Figure 12: All Loads and Supports Data (from Inventor)

Inventor-Generated Results:

$$K_{fs} = 1.405$$

$$K_f = 1.8265$$

$$\acute{\sigma}_m = \sqrt{3(K_{fs}\tau_s)^2 + 3(K_{fs}\tau_{torque})^2}$$

$$\acute{\sigma}_m = \sqrt{3(1.405 \times 1.639)^2 + 3(1.405 \times 9.623)^2} = 23.755\ Mpa$$

Length	L	226.000 mm
Mass	Mass	0.585 kg
Maximal Bending Stress	σ_B	25.990 MPa
Maximal Shear Stress	τ_S	1.639 MPa
Maximal Torsional Stress	τ	0.000 MPa
Maximal Tension Stress	σ_T	0.000 MPa
Maximal Reduced Stress	σ_{red}	26.091 MPa
Maximal Deflection	f_{max}	52.014 microm
Angle of Twist	φ	0.00 deg

Figure 13: Inventor Generated Stresses Table

$$\acute{\sigma}_a = K_f \sigma_b$$

$$\acute{\sigma}_a = (1.8265)(25.99) = 47.471\ Mpa$$

$$n_f = \left(\frac{\acute{\sigma}_a}{S_e} + \frac{\acute{\sigma}_m}{S_{ut}}\right)^{-1}$$

$$\boldsymbol{n_f = \left(\frac{47.471}{125.71} + \frac{23.755}{1041}\right)^{-1} = 2.497}$$

- This calculated n_f is very close to the given n_f value (2.5). With a minimal error percentage (0.12%), the calculations and inventor steps can be considered accurate.

$$\acute{\sigma}_{max} = \sqrt{\left(K_f \sigma_{bending}\right)^2 + 3\left(K_{fs}\left(\tau_{torque} + \tau_{shear}\right)\right)^2}$$

$$= \sqrt{(1.405 \times 25.99)^2 + 3\left(1.8265(9.623 + 1.639)\right)^2}$$

$$= 51.018\ \text{MPa}$$

$$\boldsymbol{n_y = \frac{S_y}{\acute{\sigma}_{max}} = \frac{910}{51.018} = 17.84}$$

Comparing the shaft safety factors obtained using hand calculations with the inventor-calculated values:

- Fatigue factor of safety: This calculated n_f is very close to the given n_f value (2.5). With a minimal error percentage (0.12%), the calculations and inventor steps can be considered accurate.
- Yielding factor of safety: comparing the hand-calculated value ($\mathbf{17.193}$) with the inventor-calculated value ($\mathbf{17.84}$), the error percentage is ($\mathbf{3.76}\%$). The small error percentage indicate accurate steps and calculations.

Keys Mechanical Design

The material for the Key is Aluminum wrought 2017

S_y=70, S_u =179, S_f =90

Shear yield strength:

$$S_{sy} = 0.577 S_y$$

$$S_{sy} = 0.577(70 \times 10^6) = 40390000 Pa$$

Shear force acting on the key:

$$F = \frac{T}{r}$$

$$F = \frac{15.1154}{0.009} = 1675.5\ N$$

Shear stress acting on the key:

$$\tau = \frac{S_{sy}}{n_f} = \frac{F}{tl}$$

Then:

For Gear key:

$$\frac{40390000}{n_f} = \frac{1675.5}{(0.006)(0.037)}$$

So, the safety factor for the Gear:

$$n_f = 5.35$$

For pulley key:

$$\frac{40390000}{n_f} = \frac{1675.5}{(0.006)(0.042)}$$

So, the safety factor for the pulley:

$$n_f = 6.07$$

Bearings Mechanical Design

Hand-calculated Bearings Life

For the bearing Load Life:

$$C_{10} = F_R = F_D(\frac{L_D}{L_R})^{\frac{1}{a}} = F_D[\frac{\ell_D n_D 60}{\ell_R n_R 60}]^{\frac{1}{a}}$$

From the catalog: $C_{10} = 13.5,$ *as the bearings are* ball bearings a=3

Givens:

n_D*=1150 rpm*

And we can sub $\ell_R n_R 60 = 10^6$ *and* F_D *is the resultant force for the reactions in X-Y and X-Z*

Then:

For the first bearing:

$$F_{D1} = \sqrt{\left(R_{b1y}\right)^2 + (R_{b1z})^2}$$

$$F_{D1} = \sqrt{(571.9)^2 + (131.699)^2} = 586.87\ N$$

$$13.5 = \frac{586.87}{1000}[\frac{\ell_D (1150)60}{10^6}]^{\frac{1}{3}}$$

Then:

$$\ell_D = 176411.3\ h$$

For the Second bearing:

$$F_{D2} = \sqrt{\left(R_{b2y}\right)^2 + (R_{b2z})^2}$$

$$F_{D2} = \sqrt{(44.51)^2 + (434.0105)^2} = 436.29\ N$$

$$13.5 = \frac{436.29}{1000}[\frac{\ell_D (1150)60}{10^6}]^{\frac{1}{3}}$$

Then:

$$\ell_D = 429364.8\ h$$

Inventor-Generated Bearings Life

The First Bearing: the given parameters were the following:

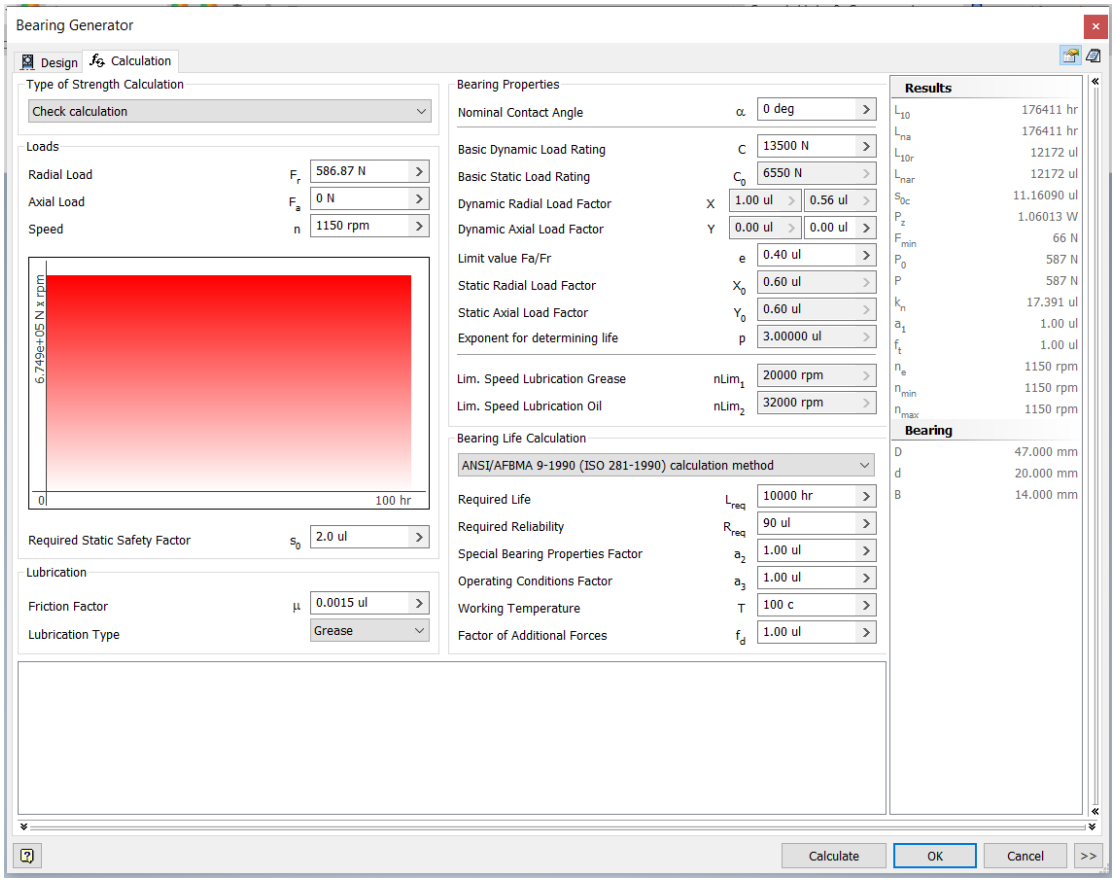


Figure 14: Inventor First Beating Given Parameters

Hence, the inventor-calculated estimated life was as the following:

$$L_{10} = 176411 \text{ hr}$$

Loads

Bearing radial load	F_r	587 N
Bearing axial load	F_a	0 N
Speed	n	1150 rpm
Required static safety factor	s_0	2.0 ul

Figure 15: First Bearing Loads (From Inventor)

Basic rating life	L_{10}	176411 hr

Figure 16: Estimated Life Value (From Inventor)

The Second Bearing: the given parameters were the following:

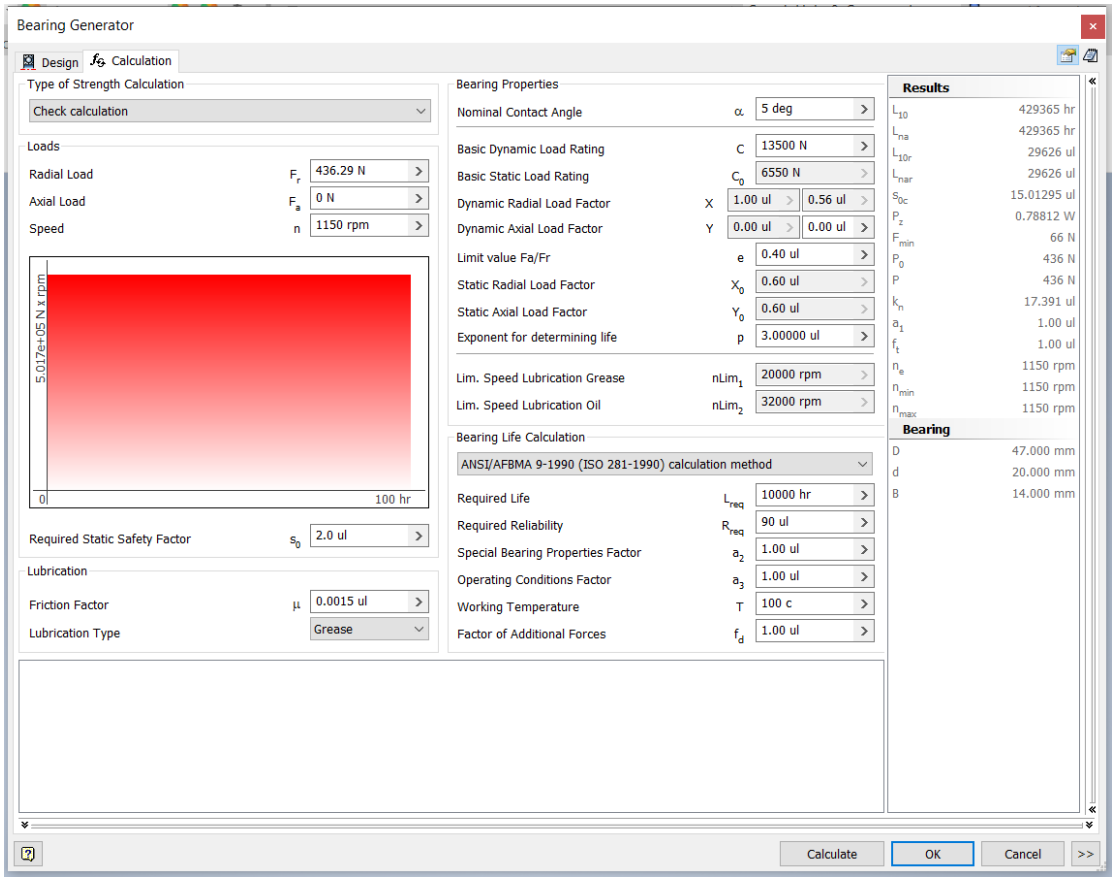


Figure 17: Inventor First Beating Given Parameters

Hence, the inventor-calculated estimated life was as the following:

$$L_{10} = 429365 \text{ hr}$$

Loads

Bearing radial load	F_r	436 N
Bearing axial load	F_a	0 N
Speed	n	1150 rpm
Required static safety factor	s_0	2.0 ul

Figure 18: First Bearing Loads (From Inventor)

Basic rating life	L_{10}	429365 hr

Figure 19: Estimated Life Value (From Inventor)

Results Comparison:

- **The first bearing:** It can be seen that the inventor-calculated estimated life value and the hand-calculated values were exact

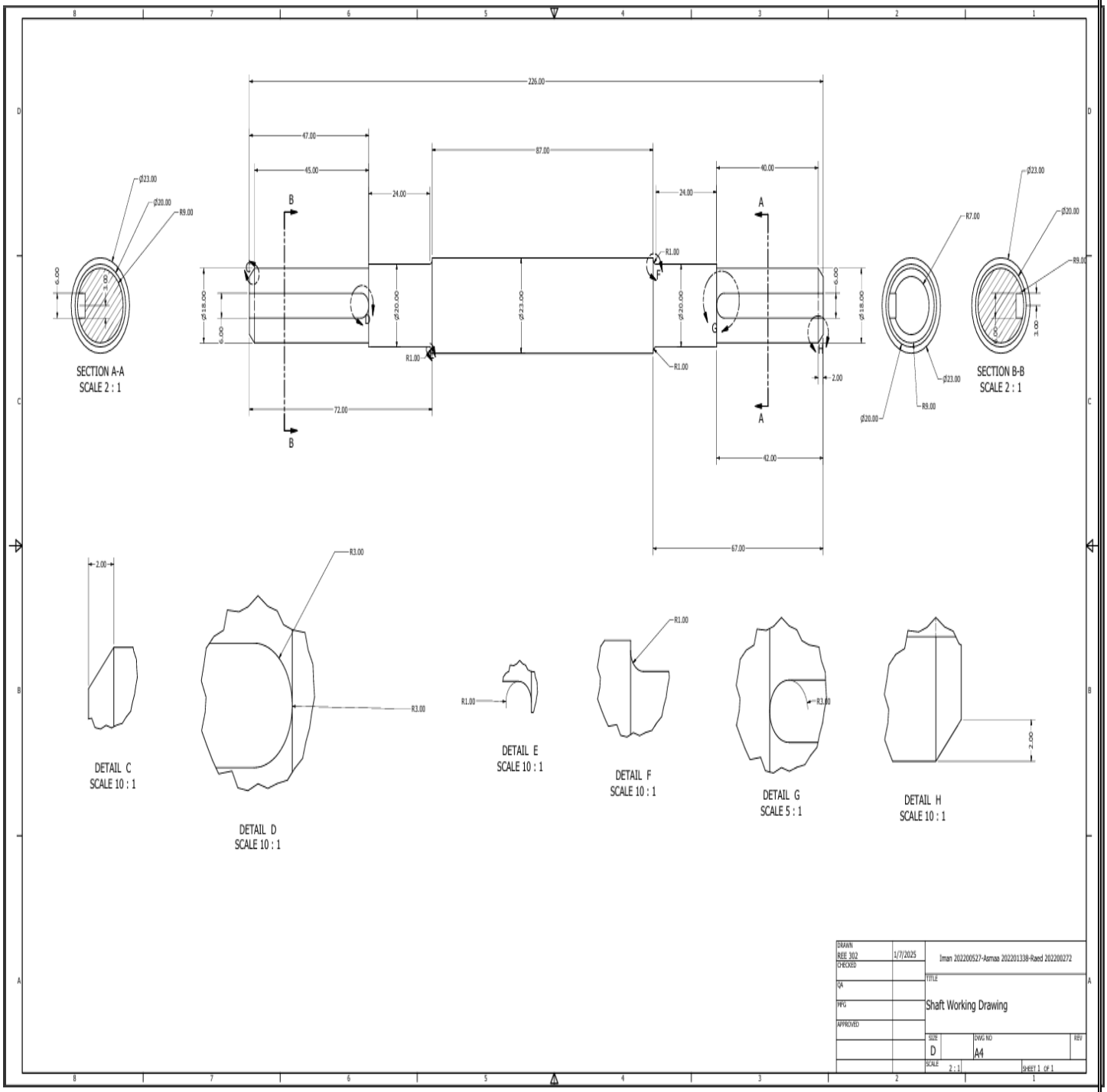
$$L_{10} = 176411 \text{ hr}$$

- **The first bearing:** It can be seen that the inventor-calculated estimated life value and the hand-calculated values were approximately equal.

$$L_{10} \approx 429365 \text{ hr}$$

- **These results indicate accurate steps and measurements.**

Shaft Working Drawing



Conclusion

The shaft was designed to withstand torsional and bending loads, with safety factors for fatigue and yielding thoroughly evaluated to prevent failure. Bearings were selected based on their ability to handle the expected radial and axial loads, and their estimated life span confirms the reliability of the system. The V-belt drive was optimized to transmit power efficiently while minimizing slippage and wear.

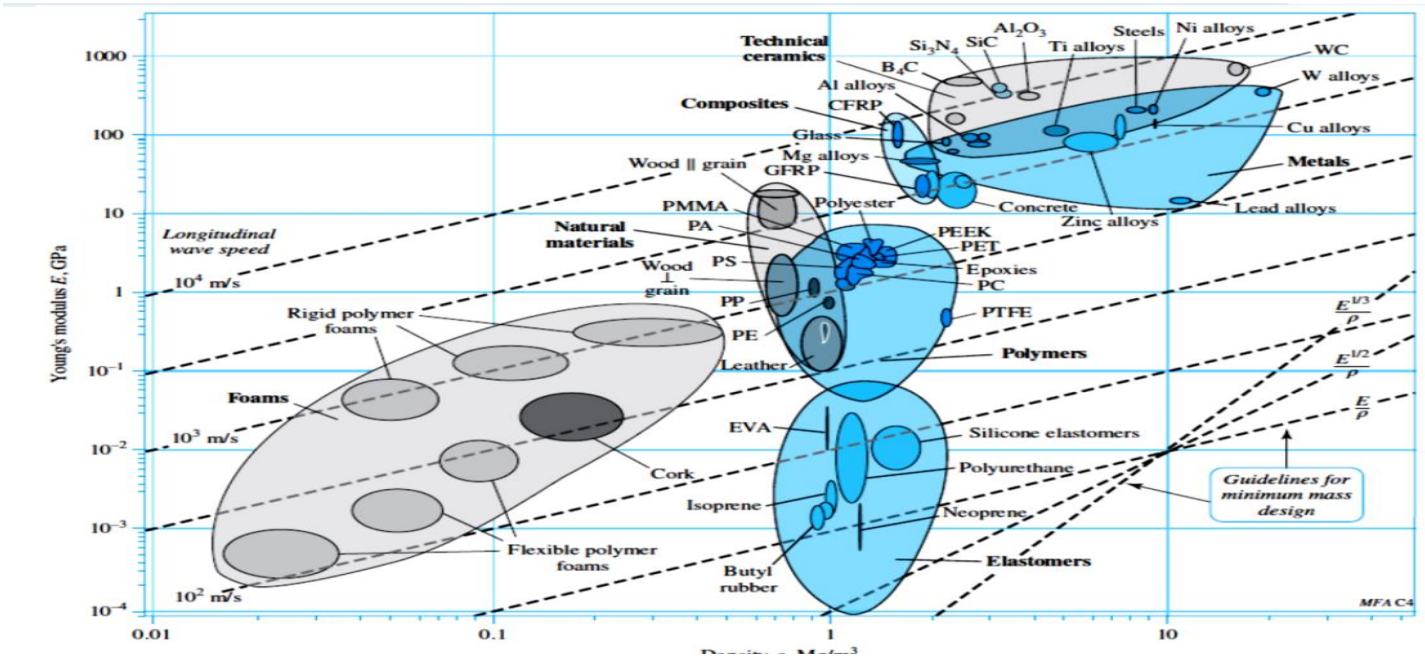
The integration of manual calculations with Autodesk Inventor modeling provided an effective balance between theoretical accuracy and practical visualization. The final assembly and component drawings, coupled with the analytical results, validate the robustness and functionality of the design.

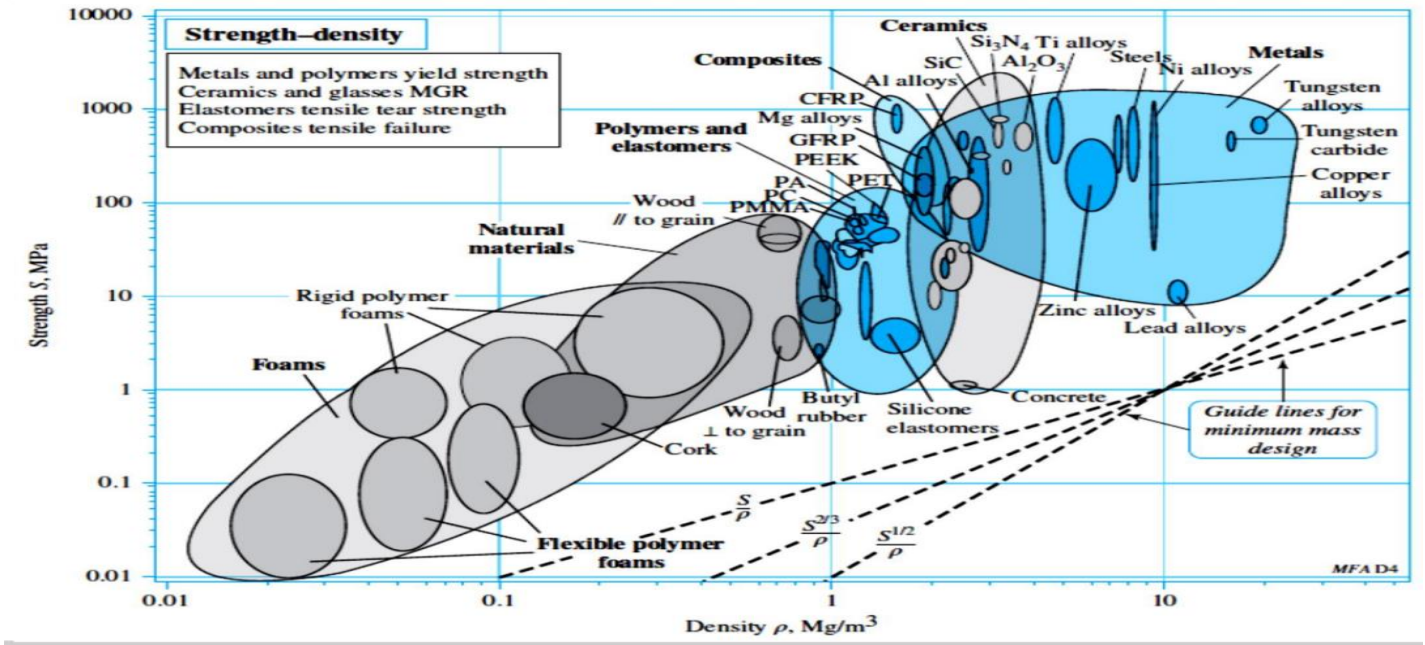
This project underscores the importance of combining engineering principles with modern tools to develop reliable and efficient mechanical systems. The outcomes align with industry standards, and the system is well suited for its intended application. Future recommendations include exploring alternative materials or components to further enhance the design's performance and cost-effectiveness.

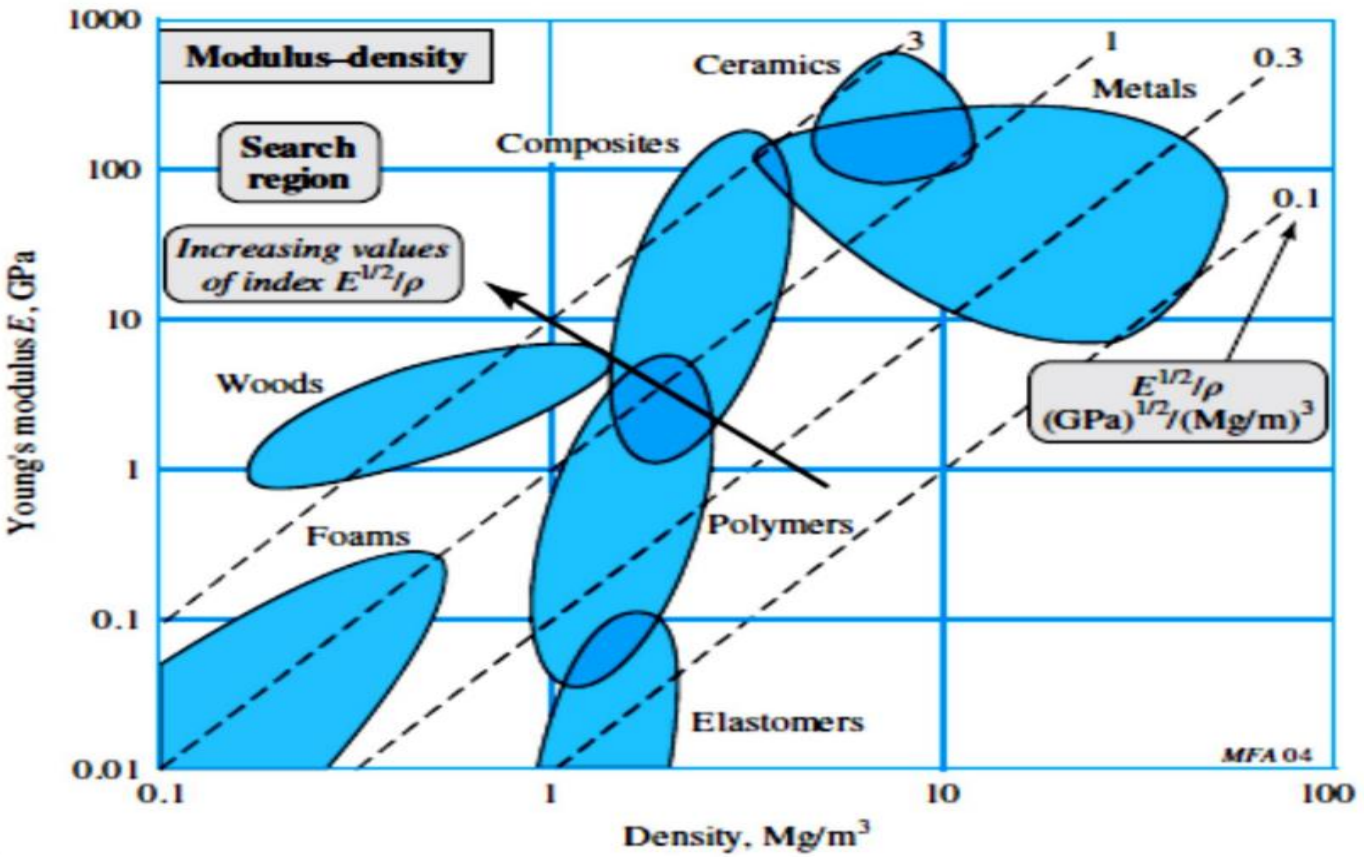
References

1. Dimarogonas, A., Author, and Lewis, , G., Reviewer (July 1, 2001). "*Machine Design: A CAD Approach*." ASME. *Appl. Mech. Rev*. July 2001; 54(4): B65–B68. https://doi.org/10.1115/1.1383679
2. *Budynas, Richard G. (Richard Gordon) author. Shigley's Mechanical Engineering Design. New York, NY :McGraw-Hill Education, 2020.*
3. Childs, Peter RN. *Mechanical design*. Elsevier, 2003.

Appendices

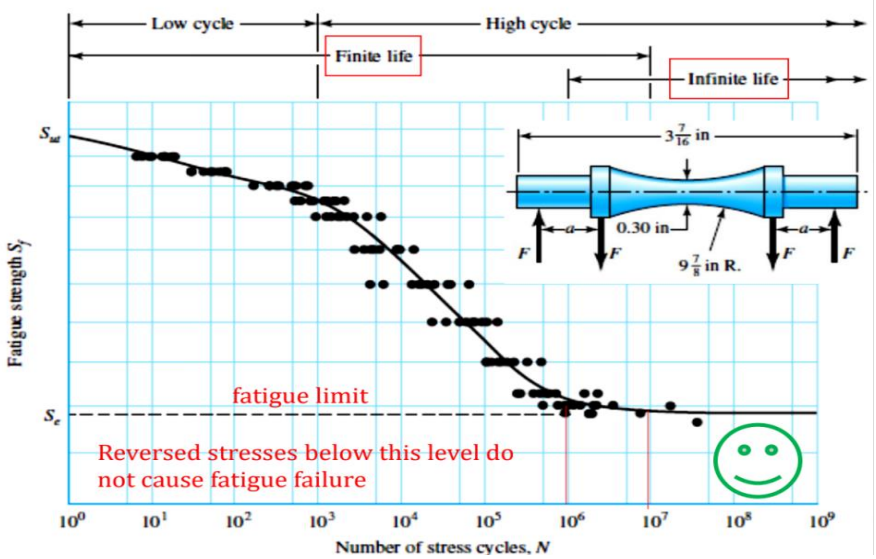

Young's modulus E, GPa
Density ρ, Mg/m3
Technical ceramics
Composites
Natural materials
Metals
Polymers
Foams
Elastomers
Longitudinal wave speed
10^4 m/s
10^3 m/s
10^2 m/s
Si3N4
SiC
Al2O3
Steels
Ni alloys
WC
W alloys
B4C
Al alloys
CFRP
Cu alloys
Glass
Mg alloys
GFRP
Wood ∥ grain
Concrete
Zinc alloys
Lead alloys
PMMA
Polyester
PA
PEEK
PET
PS
Epoxies
Wood ⊥ grain
PC
PP
PTFE
PE
Leather
Rigid polymer foams
EVA
Silicone elastomers
Polyurethane
Cork
Isoprene
Neoprene
Flexible polymer foams
Butyl rubber
Guidelines for minimum mass design
MFA C4

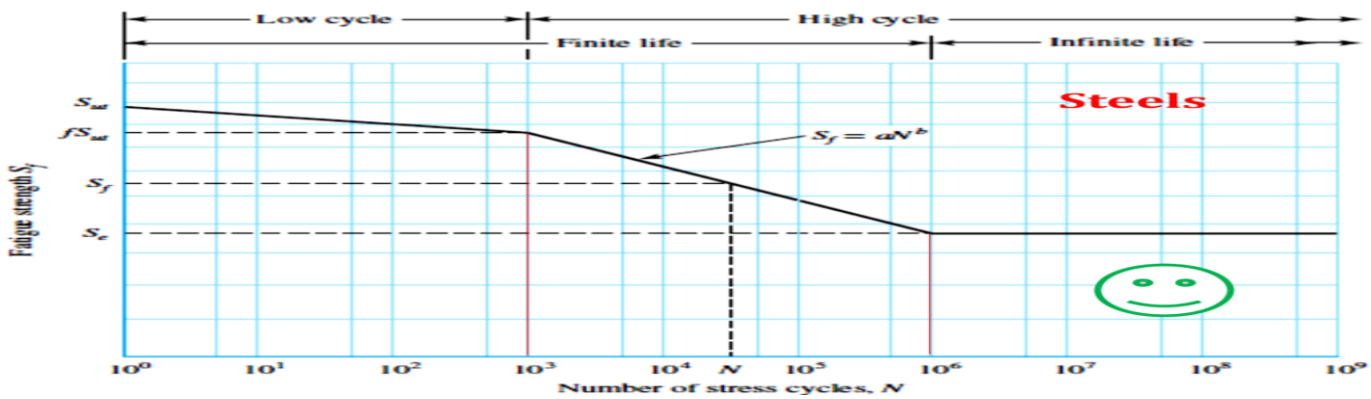

Strength–density
Metals and polymers yield strength
Ceramics and glasses MGR
Elastomers tensile tear strength
Composites tensile failure
Strength S, MPa
Density ρ, Mg/m3
Ceramics
Composites
Polymers and elastomers
Natural materials
Metals
Foams
Flexible polymer foams
Si3N4
Ti alloys
SiC
Al2O3
Steels
Ni alloys
Tungsten alloys
Tungsten carbide
Copper alloys
Al alloys
CFRP
Mg alloys
GFRP
PEEK
PA
PET
PC
PMMA
Wood // to grain
Rigid polymer foams
Zinc alloys
Lead alloys
Concrete
Wood ⊥ to grain
Butyl rubber
Silicone elastomers
Cork
Guide lines for minimum mass design
MFA D4

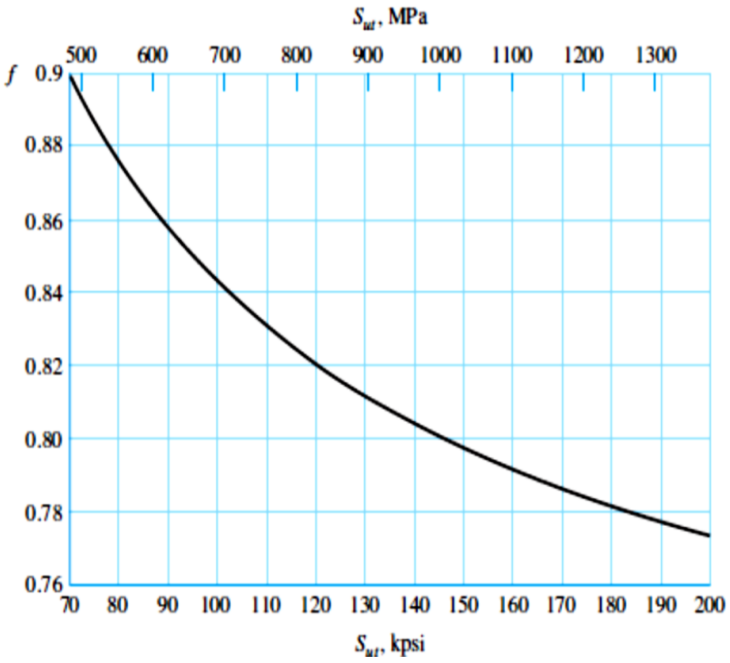


Poisson's Ratio ν

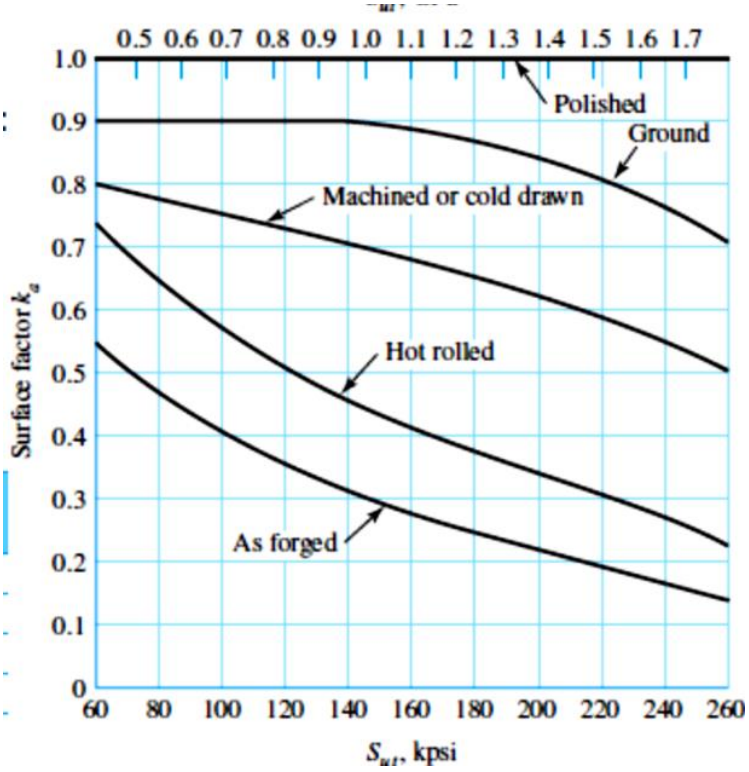
Material	ν
Aluminum	0.34
Copper	0.35
Iron	0.28
Steel	0.28
Magnesium	0.33
Titanium	0.34

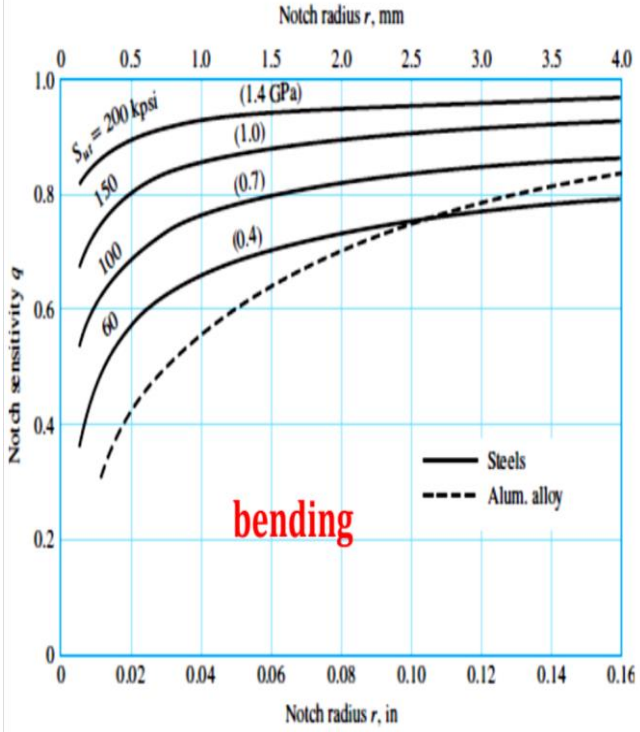




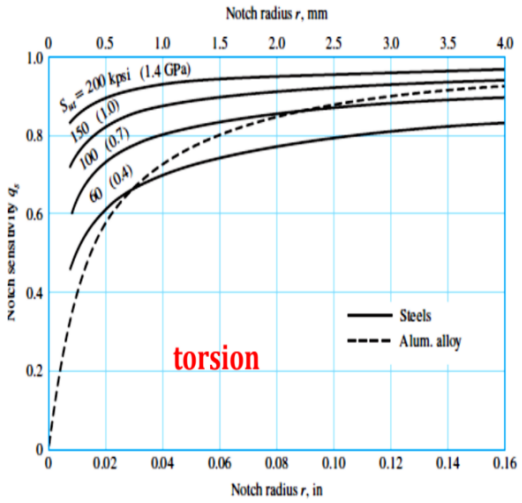


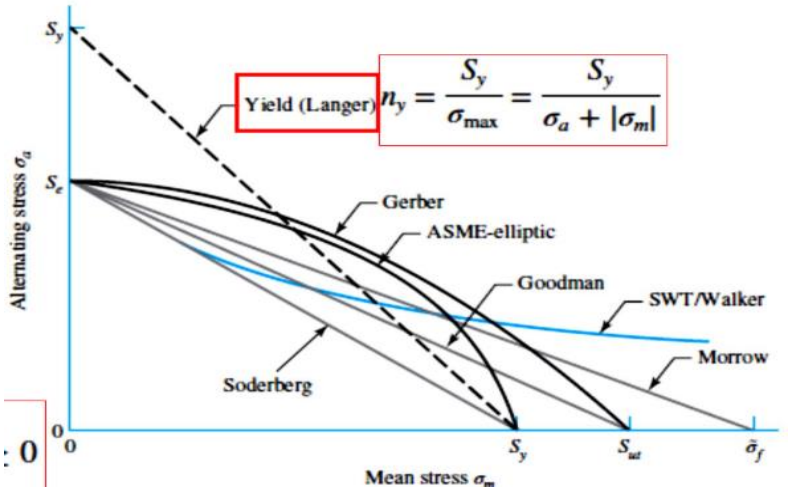
Surface Finish	Factor a		Exponent b
	S_{ut}, kpsi	S_{ut}, MPa	
Ground	1.21	1.38	-0.067
Machined or cold-drawn	2.00	3.04	-0.217
Hot-rolled	11.0	38.6	-0.650
As-forged	12.7	54.9	-0.758

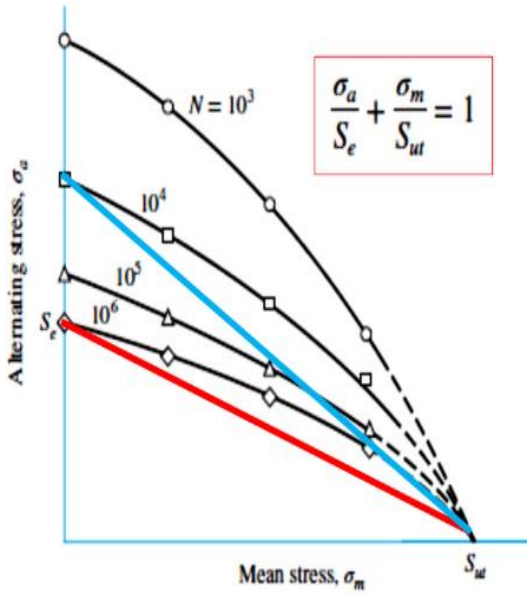




Reliability, %	Transformation Variate z_a	Reliability Factor k_e
50	0	1.000
90	1.288	0.897
95	1.645	0.868
99	2.326	0.814
99.9	3.091	0.753
99.99	3.719	0.702





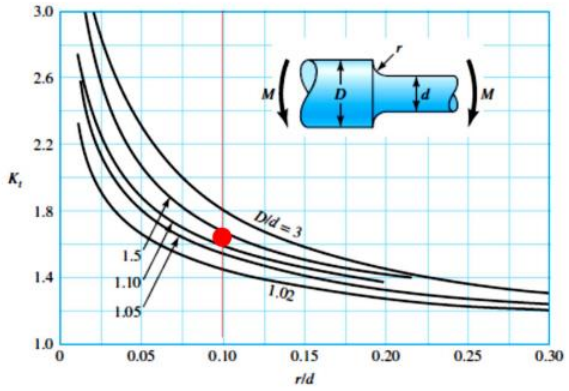


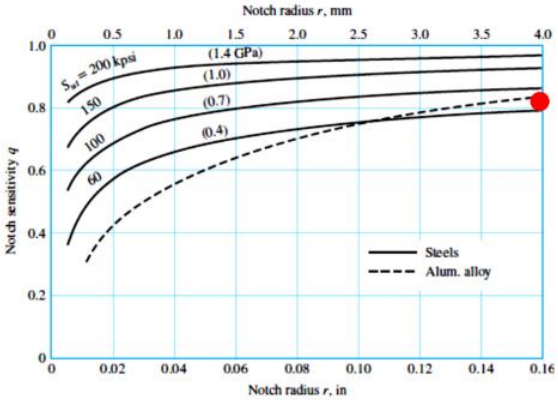
1 UNS No.	2 SAE and/or AISI No.	3 Process-ing	4 Tensile Strength, MPa (kpsi)	5 Yield Strength, MPa (kpsi)	6 Elongation in 2 in, %	7 Reduction in Area, %	8 Brinell Hardness
G10200	1020	HR	380 (55)	210 (30)	25	50	111
		CD	470 (68)	390 (57)	15	40	131
G10300	1030	HR	470 (68)	260 (37.5)	20	42	137
		CD	520 (76)	440 (64)	12	35	149
G10350	1035	HR	500 (72)	270 (39.5)	18	40	143
		CD	550 (80)	460 (67)	12	35	163
G10400	1040	HR	520 (76)	290 (42)	18	40	149
		CD	590 (85)	490 (71)	12	35	170
G10450	1045	HR	570 (82)	310 (45)	16	40	163
		CD	630 (91)	530 (77)	12	35	179
G10500	1050	HR	620 (90)	340 (49.5)	15	35	179
		CD	690 (100)	580 (84)	10	30	197

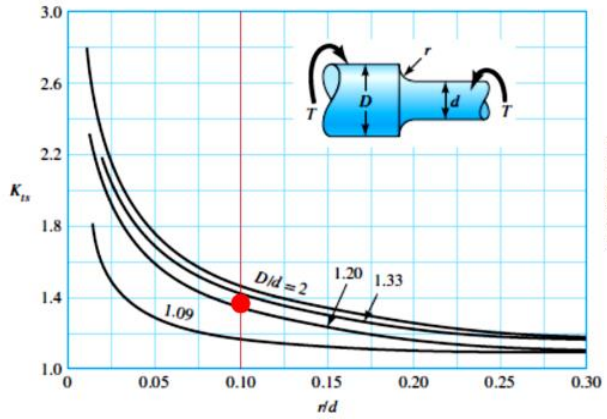
	Bending	Torsional	Axial
Shoulder fillet—sharp ($r/d = 0.02$)	2.7	2.2	3.0
Shoulder fillet—well rounded ($r/d = 0.1$)	1.7	1.5	1.9
End-mill keyseat ($r/d = 0.02$)	2.14	3.0	—
Sled runner keyseat	1.7	—	—
Retaining ring groove	5.0	3.0	5.0

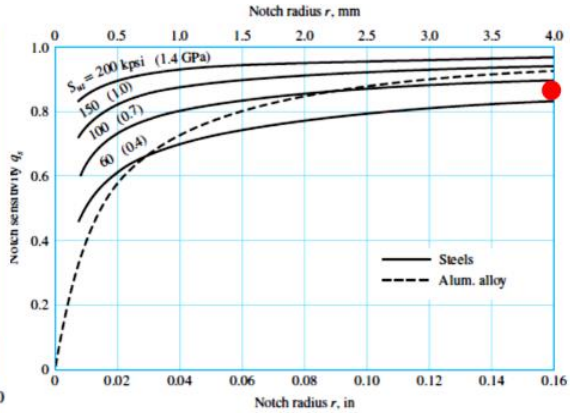
Surface Finish	Factor a		Exponent b
	S_{ut}, kpsi	S_{ut}, MPa	
Ground	1.21	1.38	−0.067
Machined or cold-drawn	2.00	3.04	−0.217
Hot-rolled	11.0	38.6	−0.650
As-forged	12.7	54.9	−0.758

	Bending	Torsional	Axial
Shoulder fillet—sharp ($r/d = 0.02$)	2.7	2.2	3.0
Shoulder fillet—well rounded ($r/d = 0.1$)	1.7	1.5	1.9

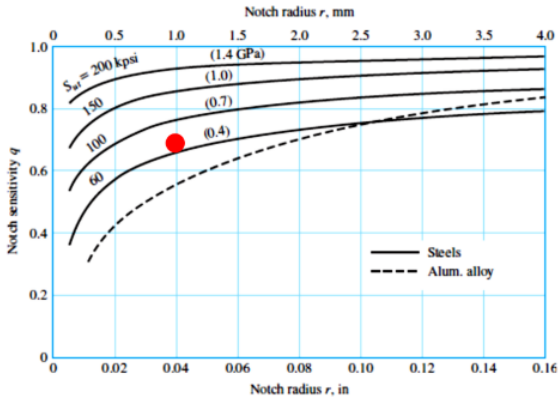


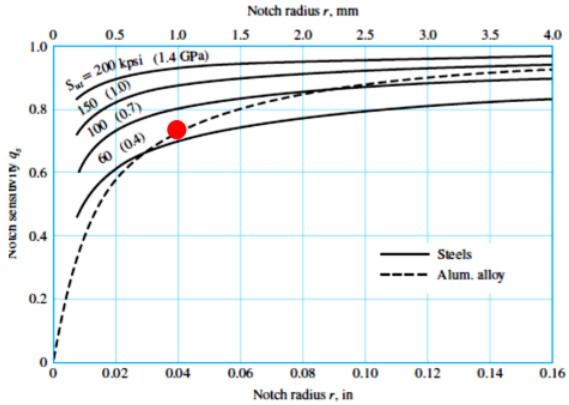


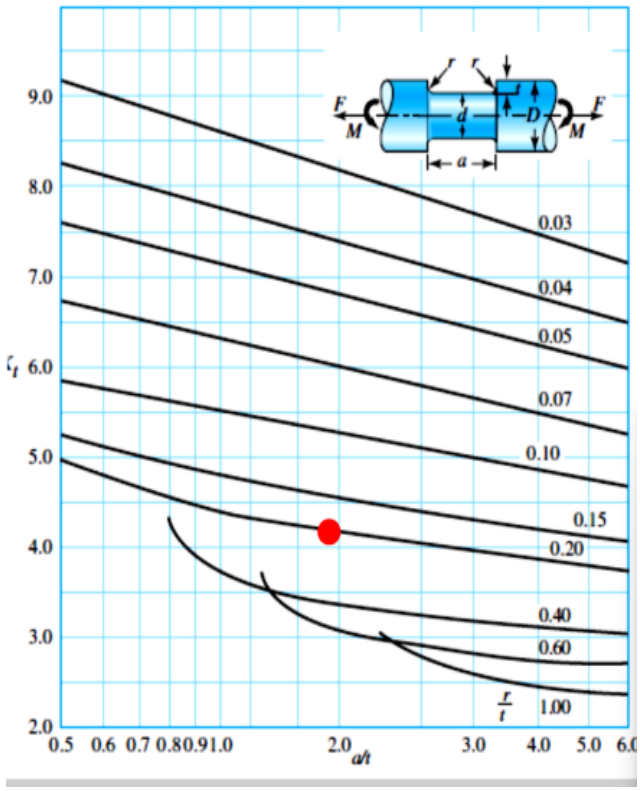




	Bending	Torsional	Axial
Shoulder fillet—sharp ($r/d = 0.02$)	2.7	2.2	3.0
Shoulder fillet—well rounded ($r/d = 0.1$)	1.7	1.5	1.9
End-mill keyseat ($r/d = 0.02$)	2.14	3.0	—







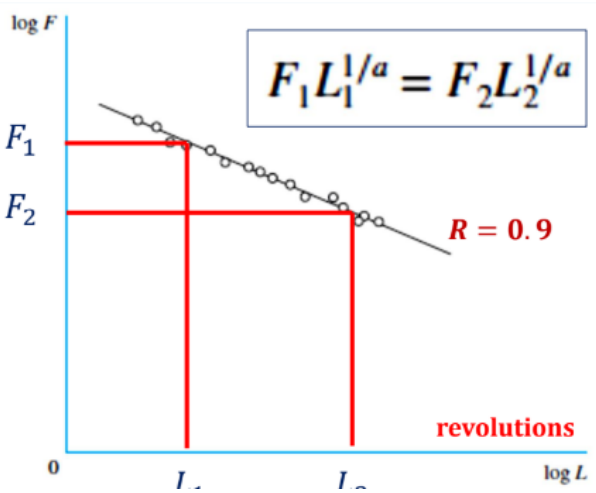
Slopes	
Tapered roller	0.0005–0.0012 rad
Cylindrical roller	0.0008–0.0012 rad
Deep-groove ball	0.001–0.003 rad
Spherical ball	0.026–0.052 rad
Self-align ball	0.026–0.052 rad
Uncrowned spur gear	<0.0005 rad

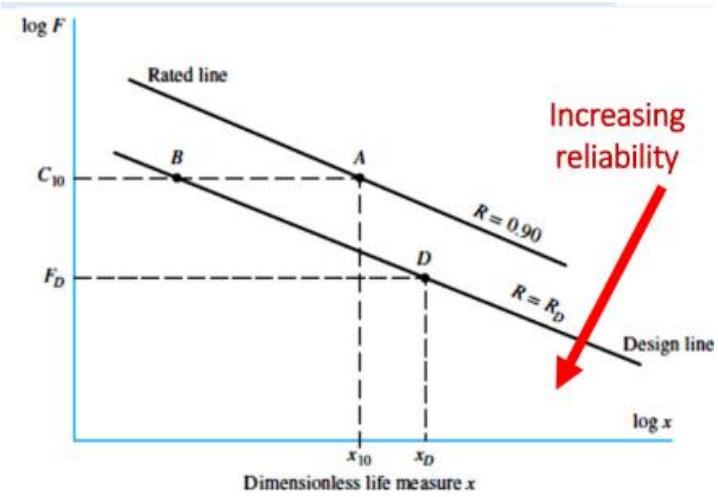
Transverse Deflections	
Spur gears with $P < 10$ Teeth/cm	0.250 mm
Spur gears with $11 < P < 19$	0.125 mm
Spur gears with $20 < P < 50$	0.075 mm

Size, mm	Seating Torque, N·m	Holding Power, N
#0	0.11	222
#1	0.2	289
#2	0.2	378
#3	0.5	534
#4	0.5	712
#5	1.1	890
#6	1.1	1112
#8	2.2	1713
#10	4.0	2403

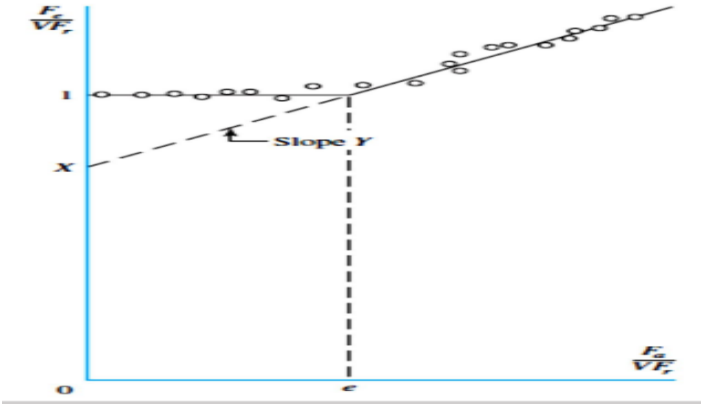
Over Shaft dia.	To Shaft dia.	Key Width	Key height	Keyway Depth
8	11	2	2	1
11	14	3 3	2 3	1 1.5
14	22	5 5	3 5	1.5 2
22	30	6 6	5 6	2 3
30	36	8 8	6 8	3 5
36	**44**	**10** **10**	**6** **10**	**3** **5**

Over Shaft dia.	To Shaft dia.	Key Width	Key height	Keyway Depth
44	58	12 12	10 12	5 6
58	70	16 16	12 16	5.5 8
70	80	20 20	12 20	6 10





		Weibull Parameters Rating Lives		
Manufacturer	Rating Life, Revolutions	x_0	θ	b
1	$90(10^6)$	0	4.48	1.5
2	$1(10^6)$	0.02	4.459	1.483



F_a/C_0	e	$F_a/(VF_r) \le e$		$F_a/(VF_r) > e$	
		X_1	Y_1	X_2	Y_2
0.014*	0.19	1.00	0	0.56	2.30
0.021	0.21	1.00	0	0.56	2.15
0.028	0.22	1.00	0	0.56	1.99
0.042	0.24	1.00	0	0.56	1.85
0.056	0.26	1.00	0	0.56	1.71
0.070	0.27	1.00	0	0.56	1.63
0.084	0.28	1.00	0	0.56	1.55
0.110	0.30	1.00	0	0.56	1.45
0.17	0.34	1.00	0	0.56	1.31
0.28	0.38	1.00	0	0.56	1.15
0.42	0.42	1.00	0	0.56	1.04
0.56	0.44	1.00	0	0.56	1.00

Bore, mm	OD, mm	Width, mm	Fillet Radius, mm	Shoulder Diameter, mm		Load Ratings, kN Deep Groove		Angular Contact	
				d_s	d_H	C_{10}	C_0	C_{10}	C_0
10	30	9	0.6	12.5	27	5.07	2.24	4.94	2.12
12	32	10	0.6	14.5	28	6.89	3.10	7.02	3.05
15	35	11	0.6	17.5	31	7.80	3.55	8.06	3.65
17	40	12	0.6	19.5	34	9.56	4.50	9.95	4.75
20	47	14	1.0	25	41	12.7	6.20	13.3	6.55
25	52	15	1.0	30	47	14.0	6.95	14.8	7.65
30	62	16	1.0	35	55	19.5	10.0	20.3	11.0
35	72	17	1.0	41	65	25.5	13.7	27.0	15.0
40	80	18	1.0	46	72	30.7	16.6	31.9	18.6
45	85	19	1.0	52	77	33.2	18.6	35.8	21.2
50	90	20	1.0	56	82	35.1	19.6	37.7	22.8

Table 11–4 Bearing-Life Recommendations for Various Classes of Machinery

Type of Application	Life, kh
Instruments and apparatus for infrequent use	Up to 0.5
Aircraft engines	0.5–2
Machines for short or intermittent operation where service interruption is of minor importance	4–8
Machines for intermittent service where reliable operation is of great importance	8–14
Machines for 8-h service that are not always fully utilized	14–20
Machines for 8-h service that are fully utilized	20–30
Machines for continuous 24-h service	50–60
Machines for continuous 24-h service where reliability is of extreme importance	100–200

Table 11–5 Load-Application Factors

Type of Application	Load Factor
Precision gearing	1.0–1.1
Commercial gearing	1.1–1.3
Applications with poor bearing seals	1.2
Machinery with no impact	1.0–1.2
Machinery with light impact	1.2–1.5
Machinery with moderate impact	1.5–3.0

									cone				cup			
bore	outside diameter	width	rating at 500 rpm for 3000 hours L_{10}		fac-tor	eff. load center	part numbers		max shaft fillet radius	width	backing shoulder diameters		max hous-ing fillet radius	width	backing shoulder diameters	
			one-row radial	thrust			cone	cup								
d	D	T	N lbf	N lbf	K	a②			R①	B	d_b	d_a	r①	C	D_b	D_a
25.000 0.9843	**52.000** 2.0472	**16.250** 0.6398	**8190** 1840	**5260** 1180	**1.56**	**–3.6** –0.14	◆30205	◆30205	**1.0** 0.04	**15.000** 0.5906	**30.5** 1.20	**29.0** 1.14	**1.0** 0.04	**13.000** 0.5118	**46.0** 1.81	**48.5** 1.91
25.000 0.9843	**52.000** 2.0472	**19.250** 0.7579	**9520** 2140	**9510** 2140	**1.00**	**–3.0** –0.12	◆32205-B	◆32205-B	**1.0** 0.04	**18.000** 0.7087	**34.0** 1.34	**31.0** 1.22	**1.0** 0.04	**15.000** 0.5906	**43.5** 1.71	**49.5** 1.95
25.000 0.9843	**52.000** 2.0472	**22.000** 0.8661	**13200** 2980	**7960** 1790	**1.66**	**–7.6** –0.30	◆33205	◆33205	**1.0** 0.04	**22.000** 0.8661	**34.0** 1.34	**30.5** 1.20	**1.0** 0.04	**18.000** 0.7087	**44.5** 1.75	**49.0** 1.93
25.000 0.9843	**62.000** 2.4409	**18.250** 0.7185	**13000** 2930	**6680** 1500	**1.95**	**–5.1** –0.20	◆30305	◆30305	**1.5** 0.06	**17.000** 0.6693	**32.5** 1.28	**30.0** 1.18	**1.5** 0.06	**15.000** 0.5906	**55.0** 2.17	**57.0** 2.24